Heart Sounds (S1, S2, S3, and S4)

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Adapted from: Heart Sounds

Listen to this Brick

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Learning Objectives (3) Versions

After completing this brick, you will be able to:

1. Explain how hemodynamic and ECG events correlate with normal S1 and S2 heart sounds.
2. Explain the physiology and clinical significance of normal and abnormal splitting.
3. Define the S3 and S4 heart sounds and describe their timing, physiology, and clinical significance.

CASE CONNECTION

MJ is a 54-year-old man with a history of poorly controlled hypertension who presents with fatigue and shortness of breath. On auscultation, you hear an additional sound right before S1—this prompts your attending to ask, "What could be causing this extra heart sound?"

As you consider the physiology behind heart sounds, think about how S3 and S4 relate to ventricular function and pathologic states. Consider your answers as you read, and we'll revisit at the end of the brick.

GO TO CONCLUSION ↓

What Are the Normal Heart Sounds (LO1)?

Cardiac auscultation is a clinical skill that involves listening to heart sounds by using a stethoscope. It is a low-tech way of detecting disease directly, without expensive scanners, blood work, or pathology reports—and another reminder that the art of medicine is not lost despite today's technology. Moreover, it's an important tool in diagnosing many forms of heart disease, a very common cause of morbidity and mortality.

So, what do you hear on auscultation of the heart? Heart sounds are short sounds produced due to a specific heart event, the closure of a valve, or the vibrating of the chordae tendineae. Before we get into pathologic sounds, we need to learn the basics of normal heart sounds. You've certainly heard the normal heart sounds before: lub-dub, lub-dub, lub-dub. Physiologists refer to these lub-dub sounds as S1 and S2 (click to hear the heart sound:) . Both of these sounds are caused by the closing of the heart valves.

Different valves open and close to allow the heart to pump blood in a coordinated fashion, and an image of the heart and the relevant valves are shown in Figure 1.

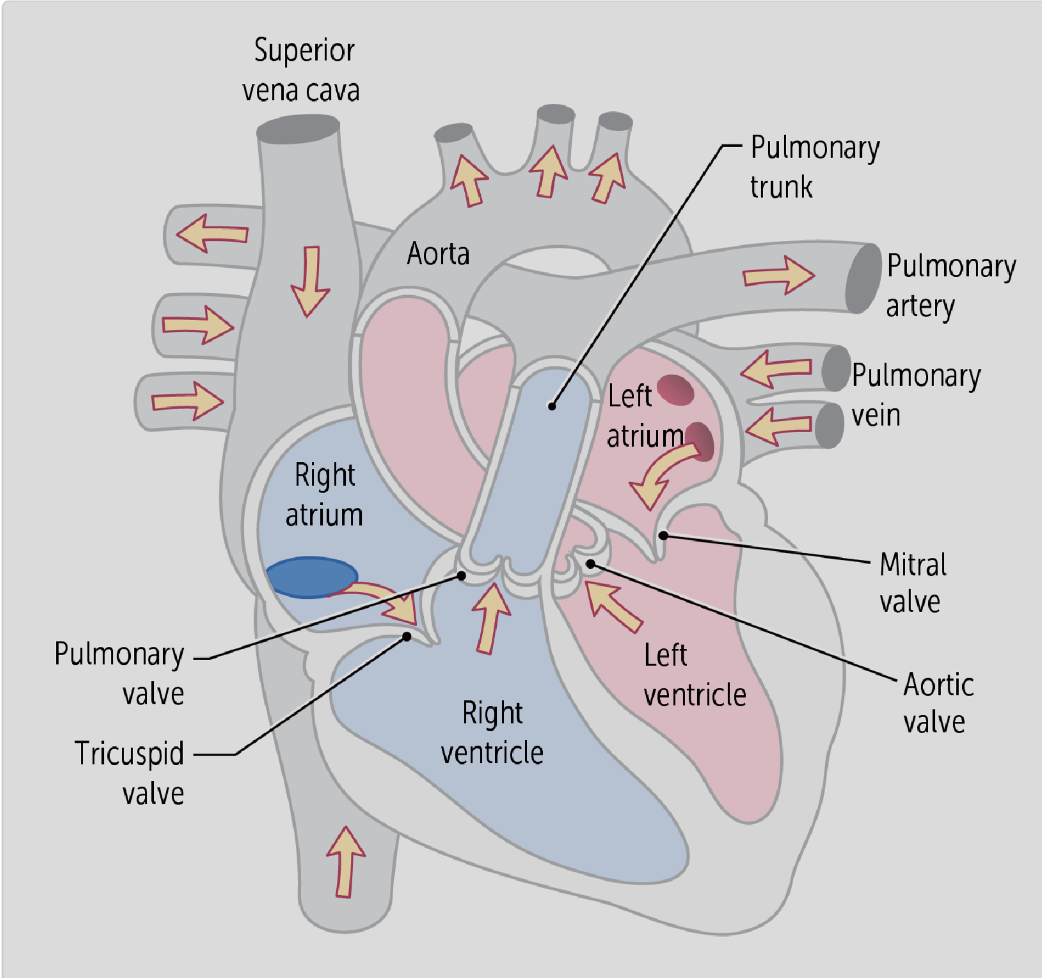

Figure 1. Location of the heart valves involved in heart sounds S1 (av valves: tricuspid/mitral valves) and S2 (pulmonary/aortic valves)

The cardiac cycle (Wiggers diagram) relates the hearts activity to the heart sounds (Figure 2). As shown on the phonocardiogram tracing on the figure, there are four possible heart sounds (called S1, S2, S3, and S4). The first two sounds, S1 and S2 (click to hear the heart sound:) , are normal sounds.

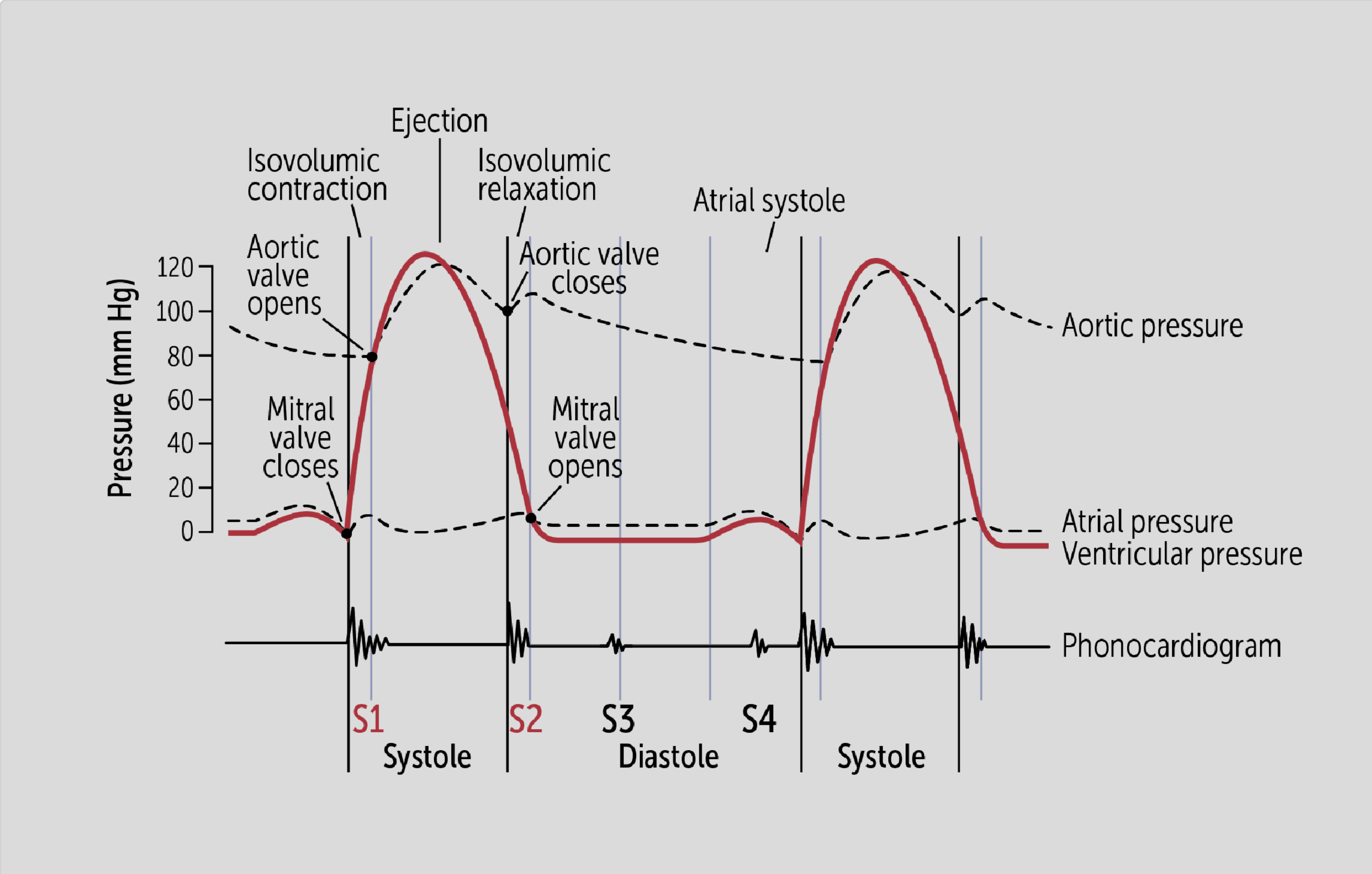

Figure 2. Heart sounds depicted on the Cardiac cycle or Wiggers diagram

The timing of heart sounds is closely linked to ventricular electrical activity on the ECG. Importantly, heart sounds arise from pressure changes following electrical events—not from the electrical activity itself

- **S1** occurs just after the QRS complex, when ventricular depolarization initiates isovolumic contraction and rising ventricular pressure closes the mitral and tricuspid valves, while
- **S2** occurs near the end of the T wave, when ventricular repolarization leads to relaxation and falling ventricular pressure closes the aortic and pulmonic valves.
- In diastole, **S3** occurs just after S2 during rapid ventricular filling, whereas
- **S4** occurs just before S1 during atrial contraction against a stiff ventricle;

S1 Heart Sound

The S1 heart sound occurs with the **normal** closing of the tricuspid and mitral valves. These **atrioventricular valves** separate the atria and the ventricles. They open during diastole so that the relaxing ventricles can fill with blood from the atria. When the ventricles contract during **systole**, the pressure in the ventricles rapidly increases (Figure 2). The increase in pressure pushes the atrioventricular valves closed, resulting in the first heart sound, or "lub" or the S1 sound. Since the mitral and tricuspid valves normally close nearly simultaneously, they are usually only heard as a single heart sound.

S2 Heart Sound

The S2 heart sound occurs with the closing of the **aortic and pulmonary valves** (Figure 2). When the ventricles contract during systole, the pressure built up opens the aortic and pulmonary valves, also called the semilunar valves. The contracting ventricles then expel blood through the semilunar valves by the pressure generated from contraction.

After the ventricles contract, they relax during **diastole**. Ventricular pressure becomes lower than the pressures in the pulmonary trunk and aorta. The pressure gradient briefly reverses, causing brief deceleration of forward flow and closure of the aortic and pulmonic valves, producing the second heart sound (S2) or "dub". Like S1, S2 is also due to the closure of the valves.

The tricuspid and mitral valves cause the S1 sound.

What Is Heart Sound Splitting (LO2)?

We refer to each of the heart sounds as a single event, but in actuality each is made up of two events—the closure of two different valves, one in the left heart and one in the right heart. Most of the time, the two valve closures occur so closely together that the listener hears a single sound. S1 and 2 splitting occurs because the valve closures are staggered so two distinct sounds can be heard. This is often referred to as the **physiologic splitting of S1 or S2**.

S1 splitting

Normal S1 splitting

In some normal individuals, and certain cardiac conditions, a split S1 can be heard, occurring when the mitral valve closes before the tricuspid valve. In normal people, this delay occurs upon inspiration, which delays in the closure of the tricuspid valve due to increased venous return, enhancing the splitting of the S1 sound (link to Split S1 heart sound).

Abnormal S1 splitting

A split S1 sound is common in the setting of a right bundle branch block (RBBB). A left bundle branch block has the opposite effect on S1.

Normal S2 Splitting

The S2 sound is normally heard as the closure of the aortic and pulmonary valves in close succession. We refer to these S2 sounds as A2 and P2, for the closure of the aortic valve and pulmonary valve, respectively.

During expiration, the A2 sound just barely precedes the P2 sound. When we inhale, the intrathoracic pressure—the pressure around the heart—drops. The drop in intrathoracic pressure allows more venous blood to enter the right ventricle. This means that more blood is expelled across the pulmonary valve during the next systole. The increase in blood flow across the pulmonary valve prolongs the time that the valve is open and delays the P2 sound, so it sounds after A2. So, **the S2 sound is normally split during inspiration**. Normally, the S2 sound should **not be split during expiration**.

S2 splitting is not always normal. In Figure 3, we see both normal and abnormal splitting.

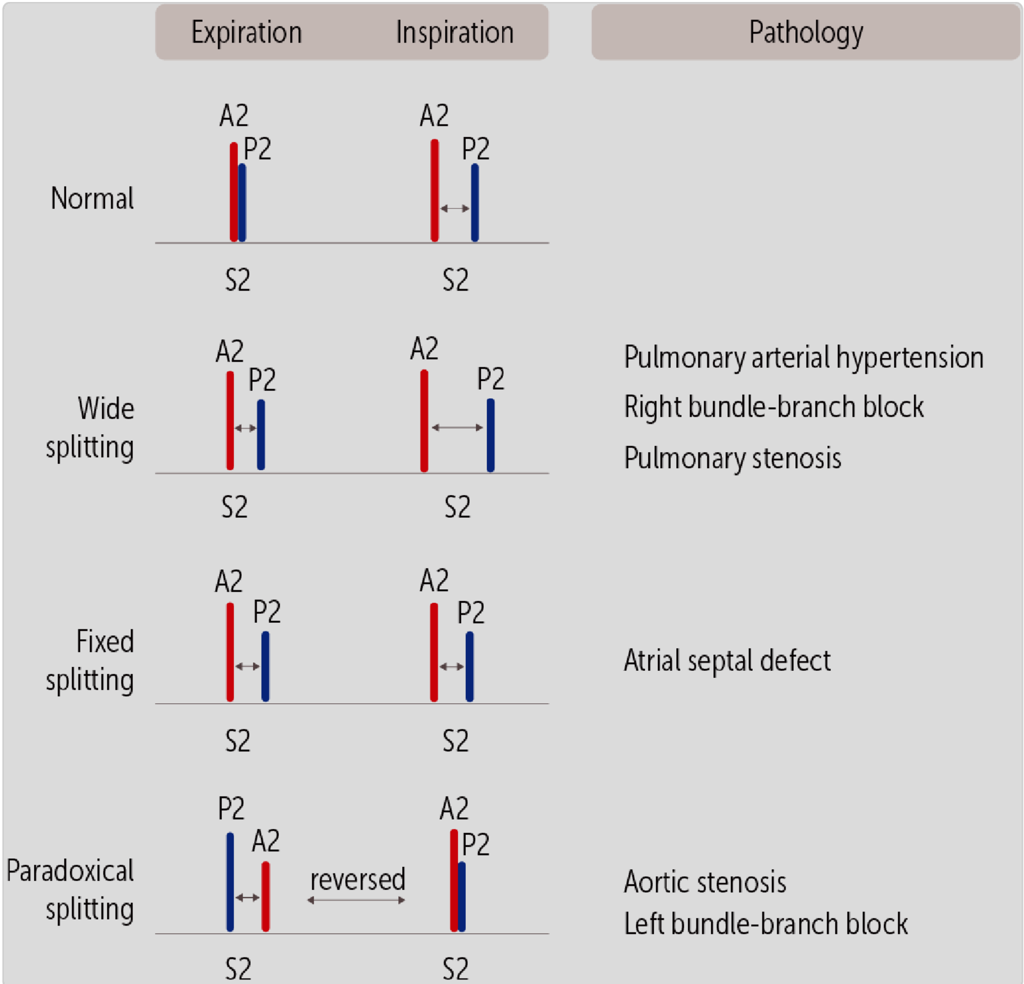

Figure 3. Visualization of the splitting of the S2 heart sound components (aortic and pulmonic; A2, P2)

The normal S2 split is heard during inspiration.

Abnormal S2 Splitting

Three types of S2 splitting are pathologic. Each is defined by its relationship to the respiratory cycle.

Wide-Split S2. In a wide-split S2, the normal S2 split is exaggerated, especially during inspiration (Figure 3). During expiration, a small split in S2 will be heard. During inspiration, a widened split will be heard, caused by the same process described for normal S2 splitting. Any condition that causes a delay in the closure of the pulmonary valve will cause a wide-split S2 (click to hear the heart sound:) ⊙. The most common examples are in the box below.

CLINICAL CORRELATION

Split S2 heart sounds:

In pulmonary arterial hypertension: high pressure in the pulmonary artery forces the **pulmonary valve to stay open longer**, delaying its closure and contributing to a **wide or fixed split S2**. Pulmonary artery hypertension is most often due to increased capillary pressure in the lung. **Pulmonary stenosis** prolongs the ejection time, also resulting in a wide split S2.

Right bundle-branch block (RBBB): delays right ventricular contraction, so the **pulmonary valve closes later**, leading to a **wider split of S2**, especially during inspiration.

False. **Physiologic splitting** of S2 occurs during inspiration due to delayed closure of the pulmonary valve. However, **pathological S2 splitting** can occur with conditions like atrial septal defect, right bundle branch block (RBBB), or pulmonary/ aortic stenosis.

Fixed-Split S2 (fixed splitting can also be called persistent splitting - Figure 3). In a fixed-split S2 (Click to hear the heart sounds:) the A2 and P2 sounds are split equally widely during BOTH expiration and inspiration, i.e., heard throughout the entire respiratory cycle (unlike wide-splitting S2, in which the split increases with inspiration). Fixed-split S2 is rare and most closely associated with the congenital heart defect **atrial septal defect** (**ASD**).

CLINICAL CORRELATION

Split S2 heart sounds:

Atrial Septal Defect: In atrial septal defect (ASD), there is an abnormal flow of blood through the heart due to a hole in the septum that separates the left atrium from the right atrium. The right atrium has lower pressure than the left atrium, so blood flows through the defect from the left atrium to the right. The extra blood in the right atrium is then pumped into the right ventricle. The right ventricle pumps the **excess blood across the pulmonary valve, delaying closure** and thus delaying pulmonary valve (P2) closure. This leads to a **fixed splitting of the**

second heart sound (S2)—heard equally during both inspiration and expiration—due to the persistent delay in P2 regardless of breathing cycle.

A fixed-split S2 is associated with atrial septal defect.

Paradoxical (Reversed) Splitting. A paradoxical split S2 heart sound is when the splitting occurs during expiration and is absent during inspiration — opposite or reversed of the physiologic split S2 (Figure 3). The aortic valve can also be a source of pathologic S2 splitting. Recall that A2 usually precedes P2 by a small amount in expiration and by a larger amount in inspiration.

CLINICAL CORRELATION

Paradoxical S2- aortic stenosis:

Aortic stenosis, the aortic valve is stiff and takes longer to close. In fact, aortic stenosis can cause enough delay in the closure of the aortic valve that A2 comes after P2 during expiration. This can be heard as a **split S2 during expiration**. Inspiration causes P2 to be delayed by the reasons we discussed before. The delay in P2 during inspiration can cause P2 to occur at the same time as the delayed A2 in aortic stenosis, causing **single S2 during inspiration**. This is the reverse of the usual pattern (single sound during expiration, split during inspiration), hence the term "paradoxical."

Left bundle-branch block (LBBB) can also be a cause of paradoxical splitting.

What Are the Third and Fourth Heart Sounds (LO3)?

In addition to the usual lub-dub of S1 and S2, additional heart sounds of **S3 and S4,** often called gallops because they resemble the gallop of a horse, can also be heard (Figure 3).

Lets take some time to differentiate between heart sounds and murmurs. Heart sounds are discrete, short audible events from a specific cause — which are different from a heart murmur. **Heart murmurs** are associated with turbulent flow, are longer with durations, even occur during the entire diastole or systole, and are typically associated with valvular diseases. The heart murmurs will be covered in their respective bricks (e.g., aortic/ mitral regurgitation, where blood flows back across a valve, or aortic/mitral stenosis- where blood flows across a stiffened valve, as well as congenital defects such as patent ductus arteriosus).

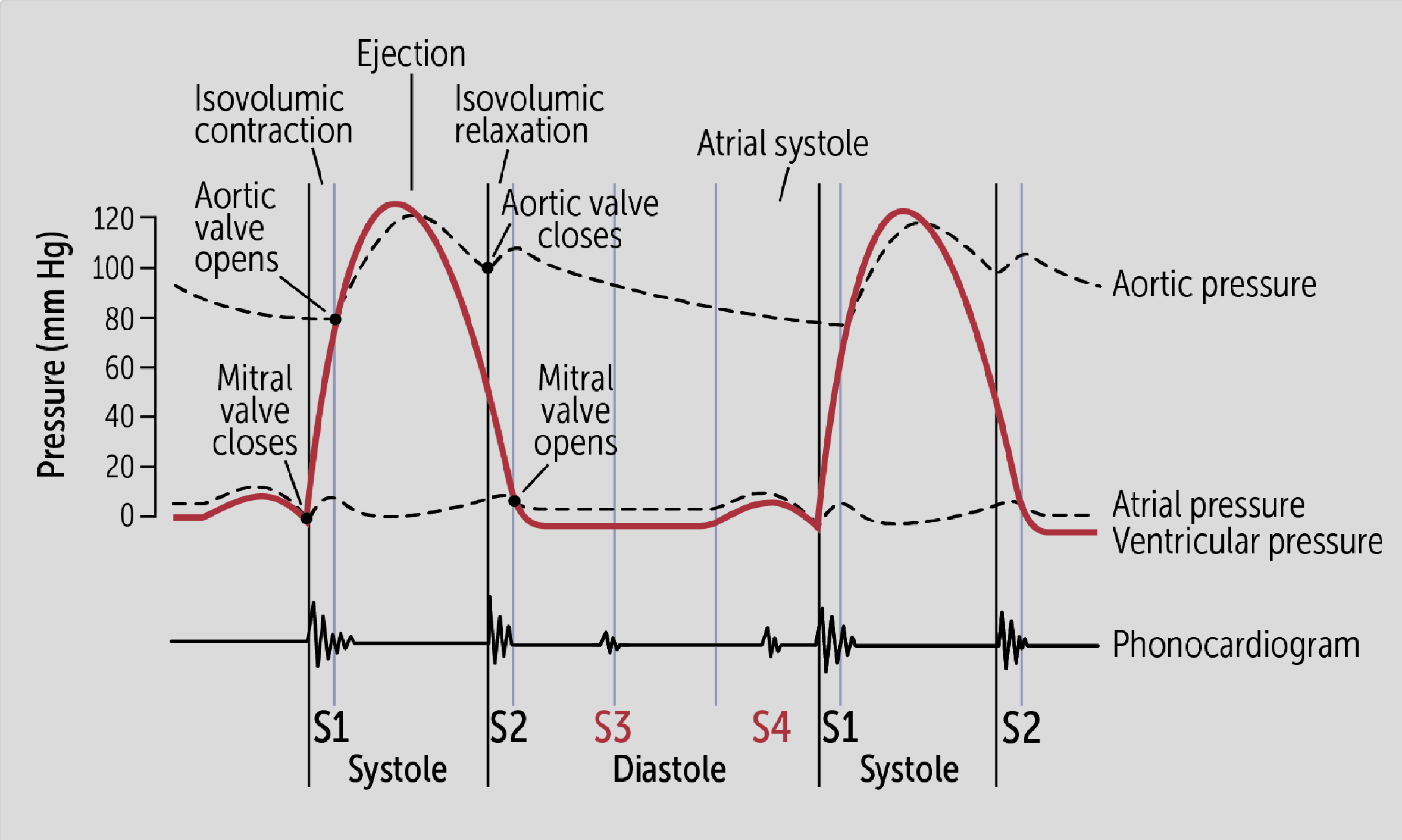

Figure 4. Relating the S3 and S4 heart sounds to the Cardiac cycle or Wiggers diagram

S3 Heart Sound

S3 is an extra sound that occurs immediately after S2. You'll need to listen to recordings to fully appreciate S3 (Click to hear the heart sounds:). Use the word "Kentucky" to help you understand the cadence of the S3, which comes immediately after the S2 (S1 is "Ken," S2 is "tuc," and S3 is "ky"):

> Ken ... tuc-ky ... Ken ... tuc-ky

Why does S3 occur? Normally during ventricular **diastole**, ventricular filling is inaudible. But sometimes S3 can be heard early in the **rapid ventricular filling** phase of the cardiac cycle. It can occur normally or pathologically. In a normal child or in a well-conditioned athlete, this sound represents the tensing of the chordae tendineae around the AV valve during ventricular filling. S3 gallops are also normal in pregnant women and well-trained athletes who have high-cardiac output states.

The S3 heart sound, **when pathological**, occurs when a large volume of blood rapidly enters a **dilated, highly compliant left ventricle** during **early diastole of the cardiac cycle** (Figure 4). During this phase, blood flows quickly from an overfilled atrium into an enlarged ventricle, causing vibrations of the ventricular wall. This produces the characteristic low-frequency "sloshing" sound, similar to pouring water rapidly into a partially filled balloon and hearing the wall vibrate.

CLINICAL CORRELATION

Pathological S3 sounds: This abnormal sound is particularly associated with conditions causing volume overload and ventricular dilation. In **dilated cardiomyopathy**, the ventricle becomes enlarged and compliant. During diastole, the sudden inflow of blood causes a **sloshing sound against the ventricle wall**, producing an **S3 gallop**.

In **mitral regurgitation**, excess volume enters the ventricle during diastole, exaggerating early filling and leading to a **prominent S3**.

An overfilled and dilated left ventricle, characteristic of volume overload (additional blood filling the heart, increasing the end-diastolic volume - EDV), states like aortic regurgitation and dilated cardiomyopathy, might cause an S3 heart sound.

S4 Heart Sound

S4 is **generally pathologic**, occurs after S2, but later in diastole than S3. In fact, it's easy to think of S4 as coming right before S1, at the very end of diastole. Use the word "Tennessee" to help understand the cadence of S4 (Click to hear the heart sounds:), which immediately precedes S1 (S4 is "Ten," S1 is "nes," and S2 is "see").

> Ten-nes ... **see** ... Ten-nes ... **see**...

The S4 occurs during atrial contraction (**late diastolic filling,** Figure 3) when blood is actively pushed from the atria into **pathologically stiff ventricles** and causes the the **chordae tendinea of the ventricles to reverberate**. What is the underlying pathology? The walls of the ventricle become thickened, typically in response to generating high pressures to eject blood from the ventricles into the aorta during **chronic hypertension** or **aortic stenosis**. The stiff and non-compliant ventricles are less able to flex or bend (**concentric ventricular hypertrophy**). The blood forcefully ejected from the atria (atrial kick) hits the stiff ventricular walls, causing the chordee tendinea of the ventricles to reverberate and produce the S4 sound.

S4. It occurs when the atrium contracts into a stiff ventricle, often due to **chronic hypertension** or **aortic stenosis**.

Remember the cadences with "Kentucky" and "Tennessee."

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to MJ, you recognize that the additional heart sound you heard was an **S4 gallop**. Given MJ's history of **chronic hypertension**, his **left ventricle has undergone concentric hypertrophy**, leading to a stiff, noncompliant ventricle. When the atrium contracts to push blood into the ventricle during diastole, it encounters resistance, producing the **S4 sound**.

You confidently explain to your attending that S4 is associated with **left ventricular hypertrophy (LVH)** due to chronic pressure overload. Your attending nods in approval, emphasizing how auscultation provides critical clues to underlying cardiac pathology.

Summary

- The normal heart sounds are called S1 and S2.
- S1 results from the closing of the atrioventricular valves (tricuspid and mitral); S2 results from the closing of the semilunar valves (aortic and pulmonary).
- The sounds fit into the cardiac cycle as: S1 → systole → S2 → diastole.
- S2 is normally a single sound during expiration and may be split during inspiration with A2 preceding P2; this is benign.
- Wide splitting of S2 during inspiration can be caused by pulmonary arterial hypertension, right bundle-branch block, or pulmonary stenosis.
- Fixed splitting of S2 is a sign of atrial septal defect.

- Paradoxical splitting occurs when A2 is delayed until after P2 during expiration and can occur in severe aortic stenosis or with left bundle-branch block.
- S3 is an additional heart sound that occurs immediately after S2; it results from atrial blood rapidly entering an overfilled ventricle.
- S3 is common in dilated cardiomyopathy.
- S4 is an additional heart sound that occurs immediately before S1; it results from the atrial kick during atrial systole.
- S4 is always pathologic and signifies noncompliant heart walls, typically from concentric ventricular hypertrophy caused by conditions that induce chronically high afterload.

Review Questions

1. Which additional heart sound can be normal in pregnant women?

A. Fixed-split S2

B. Paradoxical splitting

C. S3

D. S4

E. Wide-splitting S2

Hide Explanation

The correct answer is S3 (C). The S3 gallop can be normal in pregnant women, athletes, and children. Incorrect answers: Fixed-split S2 is a sign of atrial septal defect (A). Paradoxical splitting is caused by aortic stenosis (B). S4 is a sign of noncompliant heart walls and is always pathologic (D). Wide-splitting S2 occurs in pulmonary arterial hypertension, pulmonary stenosis, and right bundle-branch block (E).

2. A 52-year-old male comes to the office for a follow up for his congestive heart failure. On auscultation of his heart, an S4 sound is heard. What is the most likely cause of this additional heart sound in this patient?

- **A.** Atrial septal defect
- **B.** Concentric ventricular hypertrophy
- **C.** Dilative cardiomyopathy
- **D.** Heart block
- **E.** Volume overload

Hide Explanation

The correct answer is concentric ventricular hypertrophy (B). An S4 sound comes before S1 and signifies a stiff, noncompliant ventricle that recoils against the atrial kick. This can be seen with ventricular hypertrophy. Incorrect answers: An atrial septal defect (A) causes fixed splitting, not an S4 sound. Dilative cardiomyopathy

(C), heart block (D), and volume overload (E) can give rise to the S3 sound, not the S4 sound.

3. While examining a patient, you notice that the second heart sound is split while the patient is inhaling. You have her exhale, and you then hear a single second heart sound. What sound is this and what is its significance?

A. Fixed-split S2, a sign of atrial septal defect

B. Normal splitting of S2, a normal finding

C. Paradoxical splitting, a sign of aortic stenosis

D. Wide-split S2, a sign of right bundle-branch block, pulmonary stenosis, or pulmonary arterial hypertension

Hide Explanation

The correct answer is normal splitting of S2, a normal (physiologic) finding (B). S2 is normally singular during expiration and split during inspiration. Incorrect answers: A fixed-split S2 (A) is a split that stays the same regardless of respiratory status. Paradoxical splitting (C) describes when S2 is split during expiration and singular during inspiration. A wide-split S2 (D) is a split that is present during expiration and increases during inspiration.

4. During which phase of the cardiac cycle do S1 and S2 occur?

A. S1 occurs in diastole, and S2 occurs in systole

B. S1 occurs in systole, and S2 occurs in diastole

C. Both occur in diastole

D. Both occur in systole

Hide Explanation

The correct answer is (B) S1 occurs in systole, and S2 occurs in diastole. S1 marks the beginning of systole when the atrioventricular valves (mitral and tricuspid) close. S2 marks the beginning of diastole when the semilunar valves (aortic and pulmonary) close. Incorrect answers: (A) is incorrect because S1 marks systole, not diastole. (C) and (D) are incorrect because both sounds do not occur within the same phase.

5. A patient has an S2 that remains split equally during both inspiration and expiration. What is the most likely diagnosis?

A. Aortic stenosis

B. Atrial septal defect (ASD)

C. Left bundle branch block (LBBB)

D. Pulmonary arterial hypertension

Hide Explanation

Correct answer is (A) Atrial septal defect (ASD). ASD causes a fixed split S2, meaning the split remains the same regardless of respiration due to continuous left-to-right shunting of blood. Incorrect answers: (B) Pulmonary arterial hypertension does not cause a fixed split but may contribute to wide splitting. (C) Left bundle branch block (LBBB) causes paradoxical splitting, where the split is present during expiration but disappears during inspiration. (D) Aortic stenosis affects the timing of A2 but does not cause a fixed split.

6. Which of the following conditions is most likely to produce an S3 heart sound?

A. Aortic stenosis

B. Chronic hypertension

C. ○ Mitral regurgitation

D. Left bundle branch block (LBB)

Hide Explanation

The correct answer is (C) Mitral regurgitation. S3 occurs due to volume overload, which is commonly seen in mitral regurgitation as excess blood flows back into the left ventricle. Incorrect answers: (A) Aortic stenosis is associated with S4, not S3. (B) Chronic hypertension leads to ventricular hypertrophy, which is also associated with S4. (D) Left bundle branch block alters conduction but does not directly cause an S3 heart sound.

How would you rate this Brick?

References

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ~~Explain how hemodynamic and ECG events correlate with normal S1 and S2 heart sounds.~~

02 ~~Explain the physiology and clinical significance of normal and abnormal splitting.~~

03 ~~Define the S3 and S4 heart sounds and describe their timing, physiology, and clinical significance.~~

Objective Confidence: 3/3